

Dependency Patterns — Fan-Out & Fan-In

Dependencies.py

```
from airflow.sdk import dag, task
from datetime import datetime

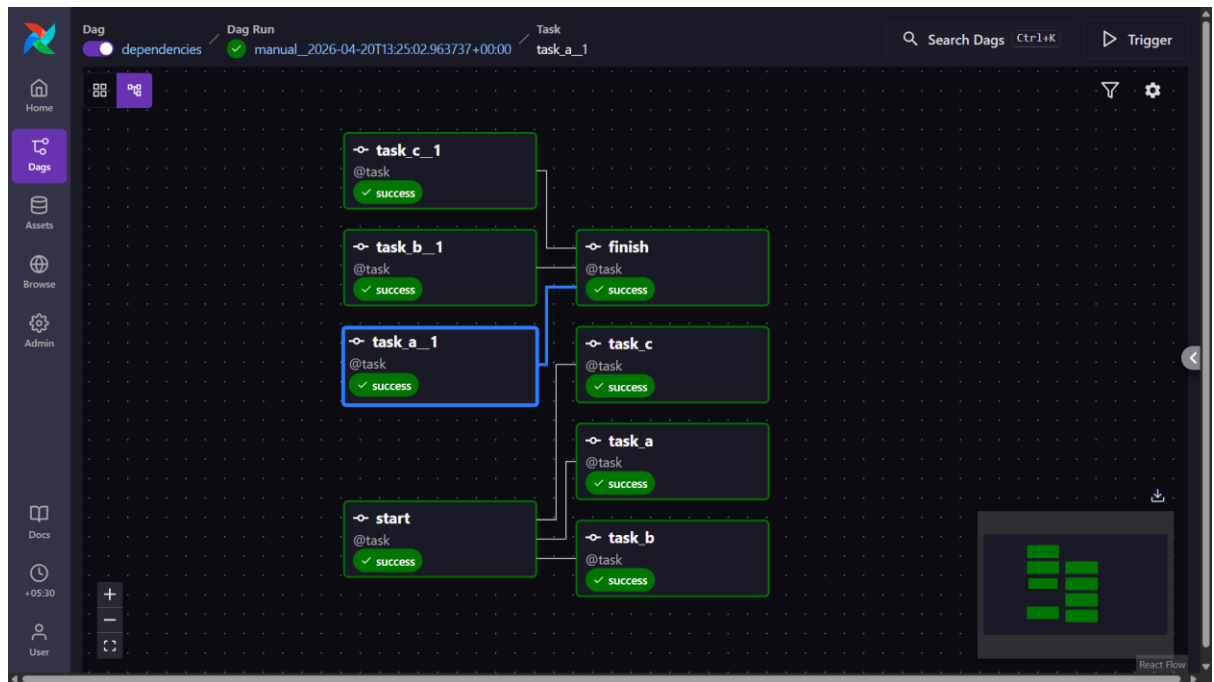
@dag(
    dag_id="dependencies",
    schedule="@daily",
    start_date=datetime(2024, 1, 1),
    catchup=False,
    tags=["Assignment-1", "dependencies"],
)
def dependencies():
    @task()
    def start(**context):
        print("Pipeline started – checking all systems")
        print(f"Execution date: {context['ds']}")

    @task()
    def task_a(**context):
        print("Task A running – processing customer data")
    @task()
    def task_b(**context):
        print("Task B running – processing order data")
    @task()
    def task_c(**context):
        print("Task C running – processing product data")

    @task()
    def finish(**context):
        print("Pipeline finished – all tasks completed")
        print(f"Execution date: {datetime.now()}")
    start() >> [task_a(), task_b(), task_c()]

    [task_a(), task_b(), task_c()] >> finish()
dependencies = dependencies()
```

1. Graph View — showing the diamond shape: start → 3 parallel tasks at same level → finish (all GREEN)



2. Task Log of finish — showing the completion message and datetime.now() timestamp

The screenshot shows the task log for the 'finish' task. The task is marked as 'Success'. The log messages are as follows:

Log Level	Message
WARNING	Astro managed secrets backend is disabled
INFO	DAG bundles loaded: dags-folder
INFO	Filling up the DagBag from /usr/local/airflow/dags/dependencies.py
INFO	TaskInstance Details: dag_id=dependencies task_id=finish dagrun_id=manual__2026-04-20T13:25:02.963737+00:00 map_in
INFO	Pipeline finished - all tasks completed
INFO	Done. Returned value was: None
INFO	Execution date: 2026-04-20 13:25:05.429044